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CHAPTER 9
FET SMALL-SIGNAL ANALYSIS

*Edisi 7
* jawaban saja

9.2 FET Small-Signal Model

1. $g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(15\,mA)}{|-5V|} = 6\,mS$

2. $g_{m0} = \frac{2\,I_{DSS}}{|V_P|} \Rightarrow |V_P| = \frac{2\,I_{DSS}}{g_{m0}} = \frac{2(12\,mA)}{10\,mS} = 2,4V$

3. $I_{DSS} = \frac{(g_{m0})(|V_P|)}{2} = \frac{5mS\,(3,5V)}{2} = 8,75\,mA$

4. $g_m = g_{m0}\left(1 - \frac{V_{GSQ}}{V_P}\right) = \frac{2(12mA)}{|-3V|}\left(1 - \frac{-1V}{-3V}\right) = 5,3\,mS$

5. $g_m = \frac{2\,I_{DSS}}{|V_P|}\left(1 - \frac{V_{GSQ}}{V_P}\right)$

$6\,mS = \frac{2\,I_{DSS}}{2,5V}\left(1 - \frac{-1V}{-2,5V}\right)$

$I_{DSS} = 12,5\,mA$

6. $g_m = g_{m0}\sqrt{\frac{I_D}{I_{DSS}}} = \frac{2\,I_{DSS}}{|V_P|}\sqrt{\frac{I_{DSS}/4}{I_{DSS}}} = \frac{2(10mA)}{5V}\sqrt{\frac{1}{4}} = 2\,mS$

7. $g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2\,(8\,mA)}{5V} = 3,2\,mS$

$g_m = g_{mo}\left(1 - \frac{V_{GSQ}}{V_P}\right) = 3,2\,mS\left(1 - \frac{V_P/4}{V_P}\right) = 3,2\,mS\left(\frac{3}{4}\right)$

$= 2,4\,mS$

8. (a) $g_m = y_{fs} = 4,5\,mS$
   (b) $r_d = \frac{1}{y_{os}} = \frac{1}{25\mu S} = 40\,k\Omega$

9. $g_m = y_{fs} = 4,5\,mS$

$r_d = \frac{1}{y_{os}} = \frac{1}{25\mu S} = 40\,k\Omega$

$Z_o = r_d = 40\,k\Omega$

$A_v(FET) = -g_m.r_d = -(4,5\,mS)(40k\Omega) = -180$

10. $A_v = -g_m.r_d \Rightarrow g_m = \frac{-A_v}{r_d} = \frac{-200}{100\,\Omega} = 2mS$

11. (a) $g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(10mA)}{5V} = 4\,mS$

(b) $g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{6,4\,mA - 3,6\,mA}{2V - 1V} = 2,8\,mS$

(c) $g_m = 4\,mS\left(1 - \frac{-1,5V}{-5V}\right) = 2,8\,mS$

(d) $g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{3,6\,mA - 1,6\,mA}{3V - 2V} = 2mS$

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(e) $g_m = 4ms\left(1 - \frac{-2,5V}{-5V}\right) = 2ms$

12 (a) $r_d = \left.\frac{\Delta V_{DS}}{\Delta I_D}\right|_{V_{GS}} = \frac{15V - 5V}{(9,1mA - 8,8mA)} = \frac{10V}{0,3mA} = 33,334k\Omega$

(b) $V_{DS} = 10\ V,\ I_D = 9mA$

$g_{m_0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(9mA)}{4V} = 4,5mS$

13 $g_m = y_{fs} = 750\mu S = 0,75mS$

$r_d = \frac{1}{y_{OS}} = \frac{1}{10\mu S} = 100k\Omega$

14 (a) $g_{mo} = \frac{2I_{DSS}}{|V_P|} = \frac{2(8mA)}{6V} = 2,67mS$

(b) $g_{m_0} = 2,67\ mS$

15 $g_m = y_{fs} = 5,6mS,\ r_d = \frac{1}{15\mu S} = 66,67k\Omega$

16 $g_m = \frac{2(10mA)}{4V}\left(1 - \frac{-2V}{-4V}\right) = 2,5\ mS$

$r_d = \frac{1}{25\mu S} = 40k\Omega$

9.3 JFET Fixed-Bias Configurasi

17 $g_m = \frac{2(10mA)}{4V}\left(1 - \frac{-1,5V}{-4V}\right) = 3,125\ mS$

$Z_i = 1M\Omega$

$Z_o = R_D // r_d = 1,8k\Omega // 40k\Omega = 1,72k\Omega$

$A_V = -g_m(R_D // r_d) = -(3,125mS)(1,72k\Omega) = -5,375$

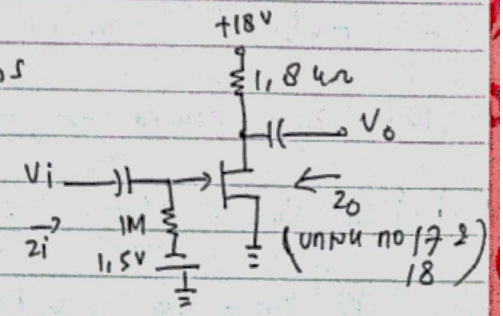
18 $g_m = \frac{2(12mA)}{6V}\left(1 - \frac{-1,5V}{-6V}\right) = 3mS$

$Z_i = 1M\Omega$

$Z_o = \frac{1}{40\mu S} = 25k\Omega$

$= 1,8k\Omega // 25k\Omega = 1,68k\Omega$

$A_V = -g_m(R_D // r_d) = -(3mS)(1,68k) = -5,04$



9.4 JFET self-bias configurasi

19 $g_m = y_{fs} = 3000\mu S = 3mS$

$r_d = \frac{1}{50\mu S} = 20k\Omega$

$Z_i = R_G = 10M\Omega$

$Z_o = 20k // 3,3k = 2,83k$

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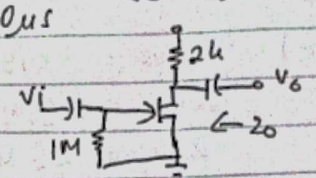
$A_v = -g_m (r_d // R_D) = -(3ms)(2,83k) = -8,49$

20 $g_{mo} = \frac{2(6mA)}{6V} = 2ms$, $r_d = \frac{1}{y_{os}} = \frac{1}{40\mu s} = 25k\Omega$

$Z_i = 1M\Omega$

$Z_o = r_d // R_D = 25k\Omega // 2k\Omega = 1,852k\Omega$

$A_v = -(2ms)(1,852k) \cong -3,7$



(no. 20 dan 47)

21 $g_m = 3ms$, $r_d = 20k\Omega$

$Z_i = 10M\Omega$

$Z_o = \frac{R_D}{1+g_m R_S + \frac{R_D+R_S}{r_d}} = \frac{3,3k\Omega}{1+(3ms)(1,1k) + \frac{3,3k+1,1k}{20k\Omega}}$

$= \frac{3,3k}{1+3,3+0,22} = \frac{3,3k\Omega}{4,52} = 730\Omega$

$A_v = \frac{-(3ms)(3,3k\Omega)}{1+(3ms)(1,1k)+\frac{3,3k+1,1k}{20k}} = \frac{-9,9}{1+3,3+0,22} = -2,19$

22 $g_m = y_{fs} = 3000\mu s = 3ms$

$r_d = \frac{1}{10\mu s} = 100k\Omega$

$Z_i = R_G = 10M\Omega$

$Z_o = r_d // R_D = 100k // 3,3k = 3,195k$

$A_v = -g_m (r_d // R_D) = -(3ms)(3,195k) = -9,59$

23 $V_{GSQ} = -0,05V$

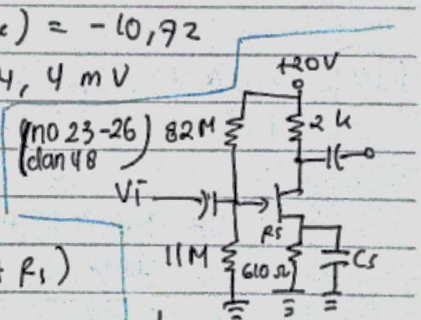
$g_m = \frac{2I_{DSS}}{V_P}\left(1 - \frac{V_{GSQ}}{V_P}\right) = \frac{2(12mA)}{3V}\left(1 - \frac{-0,05}{-3V}\right) = 5,47ms$

$Z_i = 82M\Omega // 11M\Omega = 9,7M\Omega$

$Z_o = r_d // R_D = 100k\Omega // 2k\Omega = 1,96k$

$A_v = -g_m (r_d // R_D) = -(5,47ms)(1,96k) = -10,72$

$V_o = A_v . V_i = (-10,72)(20mV) = -214,4mV$



24 $V_{GSQ} = -0,95$, $g_m = 5,47ms$

$Z_i = 9,7M\Omega$

$Z_o = \frac{R_D}{1+g_m . R_S + \frac{R_D+R_S}{r_d}}$ $\quad r_d \geq 10(R_D + R_S)$

$= \frac{2k}{1+(5,47ms)(0,61k)} = 461,1\Omega$

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$A_v = \frac{-g_m \cdot R_D}{1 + g_m \cdot R_S} = \frac{-(5{,}47\,mS)(24\,\Omega)}{4{,}337} = -2{,}52$

$V_o = A_v \cdot V_i = (-2{,}52)(20\,mV) = -50{,}40\,mV$

25 $V_{GSQ} = -0{,}95\,V$, $g_m = 5{,}47\,mS$

$Z_i = 9{,}7\,M\Omega$

$Z_o = 20\,k \,//\, 2\,k = 1{,}82\,k$

$A_v = -(5{,}47\,mS)(1{,}82\,k) = -9{,}94$

$V_o = A_v \cdot V_i = (-9{,}94)(20\,mV) = -198{,}8\,mV$

26 $V_{GS} = -0{,}95\,V$, $g_m = 5{,}47\,mS$

$Z_i = 9{,}7\,M\Omega$

$Z_o = \frac{2\,k}{1 + (5{,}47\,mS)(0{,}614) + \frac{2\,k + 0{,}614}{20\,k}} = 448{,}4\,\Omega$

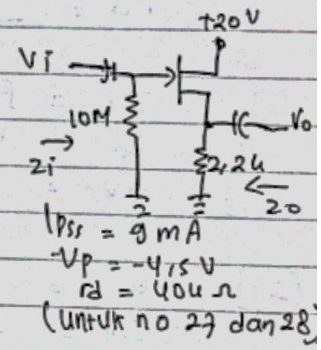
$A_v = \frac{-(5{,}47\,mS)(2\,k\Omega)}{1 + (5{,}47\,mS)(0{,}614) + \frac{2\,k + 0{,}61}{20\,k}} = -2{,}45$

27 $V_{GSQ} = -2{,}85\,V$, $g_m = \frac{2(9\,mA)}{4{,}5\,V}\left(1 - \frac{-2{,}85\,V}{-4{,}5\,V}\right) = 1{,}47\,mS$

$Z_i = R_G = 10\,M\Omega$

$Z_o = 40\,k \,//\, 2{,}24 \,//\, 1/1{,}47\,mS = 512{,}9\,\Omega$

$A_v = \frac{(1{,}47\,mS)(40\,k\Omega \,//\, 2{,}24)}{1 + (1{,}47\,mS)(40\,k \,//\, 2{,}24)} = 0{,}754$



(untuk no 27 dan 28)

28 $V_{GSQ} = -2{,}85\,V$, $g_m = 1{,}47\,mS$

$Z_i = 10\,M\Omega$

$Z_o = 20\,k \,//\, 2{,}24 \,//\, 680{,}27 = 506{,}4\,\Omega < 512{,}9\,\Omega$

$A_v = \frac{1{,}47\,mS\,(20\,k \,//\, 2{,}24)}{1 + 1{,}47\,mS\,(20\,k \,//\, 2{,}24)} = 0{,}745 < 0{,}754$

29 $V_{GSQ} = -3{,}8\,V$

$g_m = \frac{2(6\,mA)}{6\,V}\left(1 - \frac{-3{,}8\,V}{-6\,V}\right) = 0{,}733\,mS$

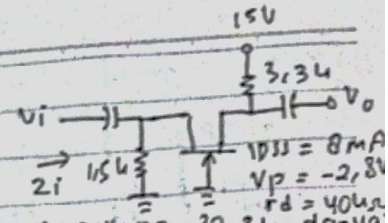
$Z_i = R_G = 10\,M\Omega$, $r_d = r_d + R_D = 30\,k + 3{,}34 = 33{,}34\,k\Omega$

$g_m = \frac{g_m \cdot r_d}{r_d + R_D} = \frac{(0{,}733\,mS)(30\,k\Omega)}{30\,k + 3{,}34} = 0{,}66\,mS$

$= 33{,}34\,k \,//\, 3{,}3k\Omega \,//\, 1/0{,}66\,mS = 1\,k\Omega$

Dipindai dengan CamScanner

$$A_v = \frac{0{,}66\,ms\,(3{,}4k\Omega)}{1 + 0{,}66\,ms\,(3{,}4k)} = 0{,}66$$



(untuk no 30, 31, dan yg

30 $V_{GSQ} = -1{,}75\,V$, $g_m = 2{,}14\,ms$

$Z_i \cong R_S \,//\, 1/g_m = 1{,}5\,k \,//\, 1/2{,}14\,ms = 356{,}3\,\Omega$

$Z_o \cong R_D = 3{,}3\,k\Omega$

$A_v \cong g_m . R_D = (2{,}14\,ms)(3{,}3k) = 7{,}06$

$V_o = A_v . V_i = 7{,}06 \times 0{,}1\,mV = 0{,}706\,mV$

31 $V_{GSQ} = -1{,}75\,V$, $g_m = \frac{2(8mA)}{2{,}8V}\left(1 - \frac{-1{,}75V}{-2{,}8V}\right) = 2{,}14\,ms$

$Z_i = R_S \,//\, \left[\frac{r_d + R_D}{1 + g_m . r_d}\right] = 1{,}5\,k\Omega \,//\, \left[\frac{25.4 + 3{,}3k}{1 + (2{,}14\,ms)(25\,k)}\right]$

$= 386{,}1\,\Omega$

$Z_o = R_D \,//\, r_d = 3{,}3k \,//\, 25k = 2{,}92k$

$A_v = \frac{(2{,}14\,ms)(3{,}3k\Omega) + 3{,}3k\Omega / 25k\Omega}{1 + 3{,}3k\Omega / 25k\Omega} = 6{,}36$

$V_o = A_v . V_i = 6{,}36 \times 0{,}1\,mV = 0{,}636\,mV$

32 $V_{GSQ} \cong -1{,}2V$, $g_m = 2{,}63\,ms$

$Z_i \cong 1k\Omega \,//\, 1/2{,}63\,ms = 1k\Omega \,//\, 380{,}2\Omega = 275{,}5\Omega$

$Z_o \cong R_D = 2{,}2k$

$A_v \cong 2{,}63\,ms \times 2{,}2k = 5{,}79$

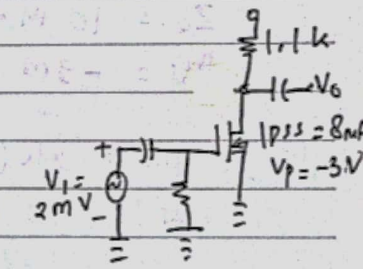
9-8 Depletion-Type MOSFET

33 $r_d = \frac{1}{20\mu S} = 50k$, $V_{GSQ} = 0V$

$g_m = \frac{2\,I_{DSS}}{V_P} = \frac{2(8\,mA)}{3} = 5{,}33\,ms$

$A_v = -(5{,}33\,ms)(1{,}1k) = -5{,}863$

$V_o = (-5{,}863)(2\,mV) = 11{,}73\,mV$

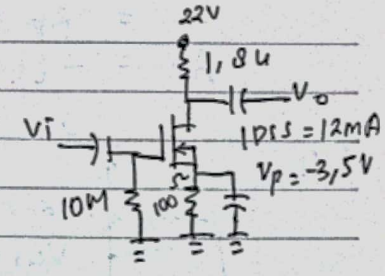


34 $V_{GSQ} = -0{,}75\,V$, $g_m = 5{,}4\,ms$

$Z_i = 10\,M\Omega$

$Z_o \cong R_D = 1{,}8\,k\Omega$

$A_v \cong -(5{,}4\,ms)(1{,}8k\Omega) = -9{,}72$



35 $Z_i = 10\,M\Omega$

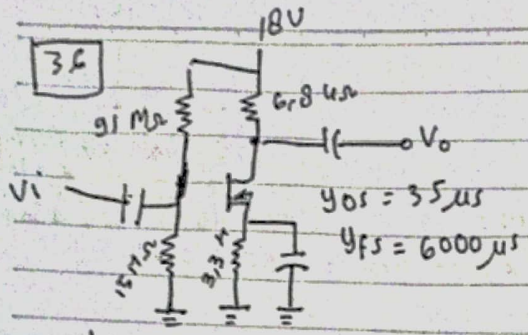
$Z_o = 25k\Omega \,//\, 1{,}8k\Omega = 1{,}68k\Omega$

$A_v = \frac{2(12\,mA)}{3{,}5\,V}\left(1 - \frac{-0{,}75V}{-3{,}5V}\right) = 5{,}4\,ms$

$A_v = -(5{,}4\,ms)(1{,}68k)$
$= -9{,}07$

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36



$g_m = y_{fs} = 6000\ \mu s = 6\ ms$

$r_d = \frac{1}{35\mu s} = 28,57\ k\Omega$

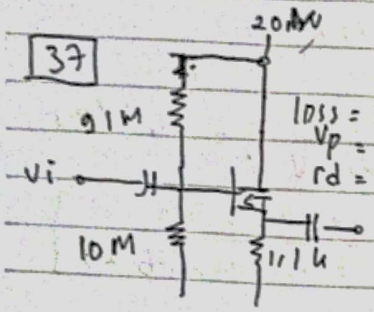
$A_v = -g_m (r_d // R_D)$

$= -(6\ ms)(28,57\ k\Omega // 6,8\ k\Omega)$

$= -32,94$

$V_o = A_v . V_i = (-32,94)(4\ mV) = -131,76\ mV$

37



$Z_i = 10\ M\Omega // 91\ M\Omega \cong 9\ M\Omega$

$g_m = \frac{2(12\ mA)}{3V}\left(1 - \frac{-1,45V}{-3V}\right) = 4,13\ ms$

$Z_o = 45\ k\Omega // 1,1\ k\Omega // 1/4,13\ ms$

$= 197,6\ \Omega$

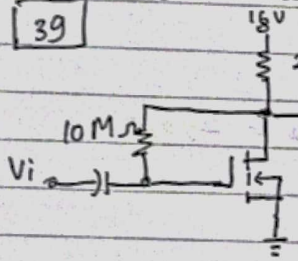
$A_v = \frac{(4,13\ ms)(45\ k\Omega // 1,1\ k\Omega)}{1 + (4,13\ ms)(45\ k\Omega // 1,1\ k\Omega)} = 0,816$

9.10 E-MOSFET DRAIN-FEEDBACK CONFIGURATION

38

$g_m = 2k(V_{GSQ} - V_{GS(Th)}) = 2(0,3 \times 10^{-3})(8V - 3V) = 3\ ms$

39



$V_{GSQ} = 6,7V$

$g_m = 2(0,3 \times 10^{-3})(6,7V - 3V) = 2,22\ ms$

$Z_i = \frac{10\ M\Omega + 100\ k\Omega // 2,2\ k\Omega}{1 + (2,22\ ms)(100\ k\Omega // 2,2\ k\Omega)}$

$= 1,73\ M\Omega$

$Z_o = 10\ M\Omega // 100\ k\Omega // 2,2\ k\Omega = 2,15\ k\Omega$

$A_v = -g_m (R_F // r_d // R_D) = -2,22\ ms\ (2,15\ k\Omega) = -4,77$

40

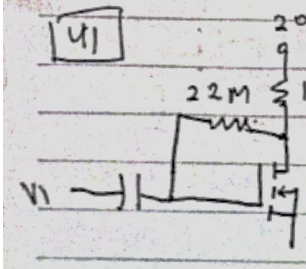
$g_m = 2k(V_{GSQ} - V_T) = 2(0,2 \times 10^{-3})(6,7V - 3V) = 1,48\ ms$

$Z_i = \frac{10\ M\Omega + 100\ k\Omega // 2,2\ k\Omega}{1 + (1,48\ ms)(100\ k\Omega // 2,2\ k\Omega)} = 2,39\ M\Omega > 1,73\ M\Omega$

$Z_o = R_F // r_d // R_D = 2,15\ k\Omega$

$A_v = -(1,48\ ms)(2,15\ k\Omega) = -3,182 < -4,77$

41



$V_{GSQ} = 5,7V$, $g_m = 2(0,3 \times 10^{-3})(5,7V - 3,5V)$

$= 1,32\ ms$

$r_d = \frac{1}{30\mu s} = 33,33\ k\Omega$

$A_v = -g_m (R_F // r_d // R_D)$

$= -1,32\ ms\ (22\ M // 33,33 // 10\ k\Omega)$

$= -10,15$

$V_o = A_v . V_i = (-10,15)(20\ mV) = -203\ mV$

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42 $k = \frac{I_D}{(V_{GS} - V_T)^2} = \frac{4mA}{(7V-4V)^2} = 0,444 \times 10^{-3}$

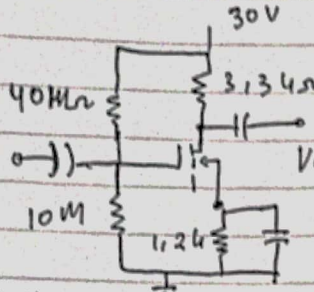
$g_m = 2(0,444 \times 10^{-3})(7V-4V) = 2,66\ mS$

$A_V = -(2,66\ mS)(22M\Omega \,//\, 50k\Omega \,//\, 10k\Omega) = -22,16$

$V_o = -22,16 \times 4mV = -88,64\ mV$

9.11 E-MOSFET Voltage-Divider Configuration

43



$V_{GS} = 3V$

$k = 0,4 \times 10^{-3}$

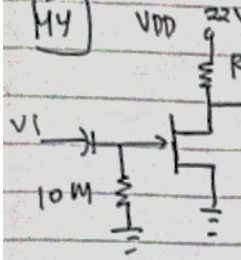
$V_{GSQ} = 2(0,4 \times 10^{-3})(4,8V-3V) = 1,44mS$

$A_V = -(1,44mS)(40k \,//\, 3,3k) = -4,39$

$V_o = (-4,39)(0,8mV) = -3,51mV$

9.12 Designing FET Amplifier Network

44



$r_d = \frac{1}{25\mu S} = 40k\Omega$

$g_{mo} = \frac{2(8mA)}{2,5V} = 6,4mS$

$|A_V| = g_m(r_d \,//\, R_D)$

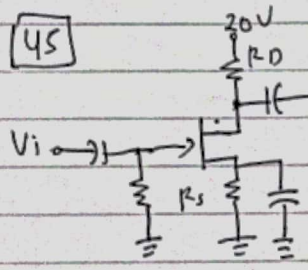
$8 = (6,4mS)(40k \,//\, R_D)$

$\frac{8}{6,4mS} = 1,25k\Omega = \frac{40k \cdot R_D}{40k + R_D}$

$R_D = 1,29k$

$R_D = 1,3k$

45



$I_{DSS} = 12\ mA$

$V_p = -3V$

$r_d = 40k$

$V_{GSQ} = \frac{1}{3}V_p = \frac{1}{3} \times (-3V) = -1V$

$I_{DQ} = 12\ mA\left(1 - \frac{-1V}{-3V}\right)^2 = 5,33\ mA$

$R_s = \frac{V_s}{I_{DQ}} = \frac{1V}{5,33mA} = 187,62\Omega$

$g_m = \frac{2(12mA)}{3V}\left(1 - \frac{-1V}{-3V}\right) = 5,33\ mS$

$A_V = -g_m(R_D \,//\, r_d) = -10$

atau $R_D \,//\, 40k\Omega = \frac{-10}{5,33mS} = 1,876k\Omega$

$\frac{R_D \cdot 40k}{R_D + 40k} = 1,876k\Omega$

$40k\Omega R_D = 1,876k \cdot R_D + 75,04k\Omega^2$

$R_D = 1,97k\Omega = 2k\Omega$

Dipindai dengan CamScanner